An Exploratory Benchmark of Speech Emotion Recognition Models for Clinical Use-Case Considerations

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Abstract

Speech-based emotion recognition (SER) has gained growing interest as a non-intrusive indicator of psychological well-being. This project explores its feasibility as a foundation for future conversational screening and mental-health monitoring tools. Using three publicly available emotional speech corpora (RAVDESS, CREMA-D, TESS), we benchmark classical models (Log Regression, SVM, XGBoost) and deep neural networks (CNN, BiLSTM, CNN-LSTM) using MFCC-based acoustic features. Stage 1 experiments on non-augmented data showed BiLSTM achieving the highest performance (70% accuracy, 0.72 macro-F1), outperforming all baseline methods. In Stage 2, the top three models were re trained with augmented audio to improve robustness; CNN-LSTM again performed best (73% accuracy, 0.75 macro-F1), particularly enhancing recall for emotions related to SSD risk, such as sadness, fear and disgust. These findings demonstrate strong technical feasibility for deployment-oriented SER and establish a modelling benchmark to support future clinical adaptation as a possible application.

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1 Introduction

1.1 Motivation & Background Information

Our team was motivated by the growing potential of speech-recognition tools to decipher human emotions — an area where misunderstandings are common and reliable detection remains challenging. Although our project does not aim to solve these issues directly, it explores how effectively machine-learning models can recognise emotions from speech and what insights this can offer for future applications. The findings serve as early groundwork for possible healthcare and clinical use cases, where emotion-aware AI systems may one day support clinicians by passively monitoring vocal cues and flagging shifts in emotional wellbeing. While our work is not ready for deployment, it contributes to a deeper understanding of what current models can achieve and how they might be adapted for future clinical integration.

1.2 Literature Review

Speech-emotion recognition (SER) can classify affective states using acoustic features such as MFCCs with reported accuracies above 70% in controlled datasets [1]. Regional initiatives including Singapore-focused auditory mental-health research highlight growing interest in AI-enabled eldercare [2]. However, most SER datasets (e.g., RAVDESS, CREMA-D, TESS) involve younger Western speakers acting exaggerated emotions, which differ from subtle SSD-related vocal cues in seniors.

1.3 Gaps & Knowledge

Existing speech-emotion-recognition (SER) frameworks often struggle with real-world variability such as differences in speaking styles, accents, and vocal characteristics. Because of these limitations, it remains unclear how well current models can actually classify emotions from speech under controlled conditions. This project therefore aims to benchmark a range of classical and deep-learning models (SVM, XGBoost, CNN, BiLSTM, and CNN–LSTM) on existing acted-speech datasets to evaluate their relative performance and limitations.

2 Dataset and Pre-Processing

2.1 Description of Dataset

Our dataset combines three publicly available emotional-speech corpora: RAVDESS, CREMA-D, and TESS, merged to increase sample size and emotional diversity.

1. RAVDESS contains audio recordings from 24 actors expressing 8 emotions (neutral, calm, happy, sad, angry, fearful, disgust, surprised) across two intensities. Only the audio-only modality (1,440 files) was used.
2. CREMA-D includes 91 actors producing 12 sentences across 6 emotions (neutral, happy, sad, angry, fearful, disgust) and four intensity levels, contributing 7,442 audio files.
3. TESS provides 2,800 recordings from two female speakers reading 200 target words expressed in 7 emotions (neutral, happy, sad, angry, fearful, disgust, surprised).

These datasets differ in structure, speaker demographics, and recording styles, providing a varied but heterogeneous source for benchmarking SER models.

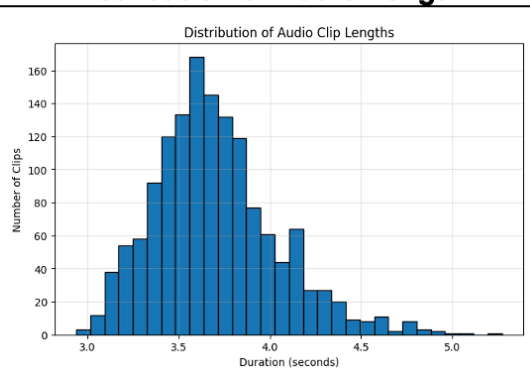
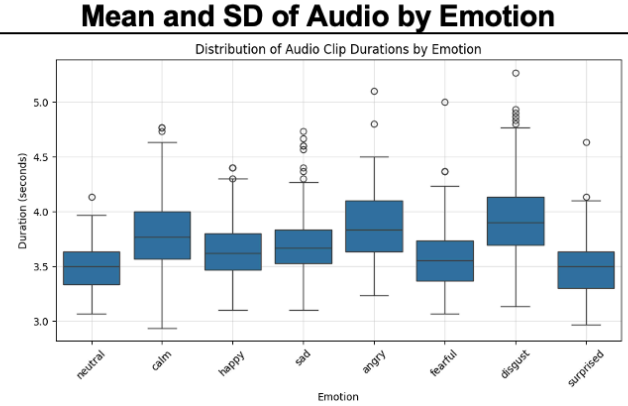
2.2 Exploratory Data Analysis

RAVDESS: Due to the structured nature of this dataset, the data is quite well organised and balanced, with only the neutral tone audio having fewer recordings, resulting in the following counts of each emotion:

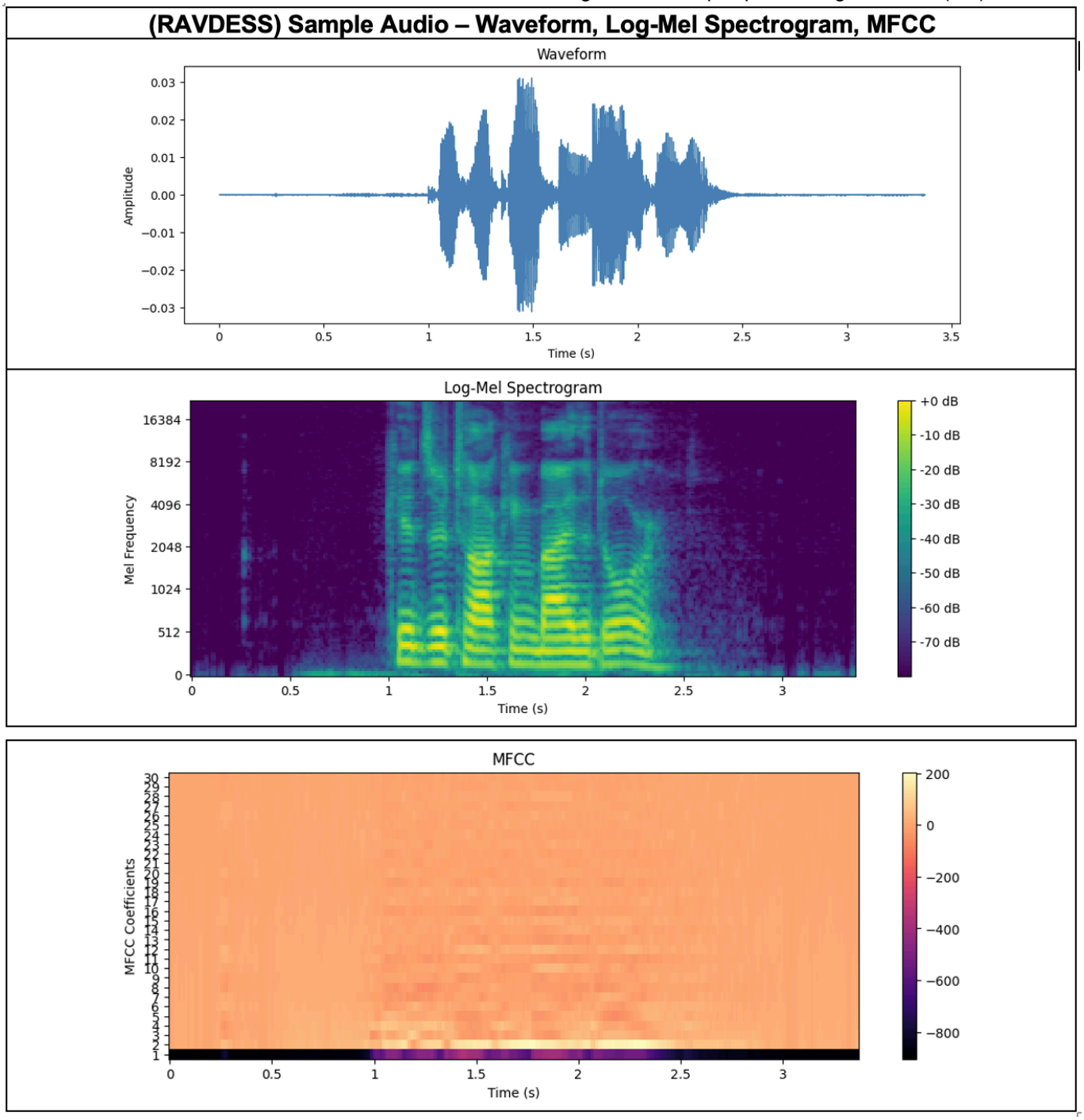
	Count of Each Emotion							
Emotions	Neutral	Calm	Happy	Sad	Angry	Fearful	Disgust	Surprised
Counts	96	192	192	192	192	192	192	192

Table 2.2.1 RAVDESS Emotion Counts

Plotting the duration distribution also shows us that most of the audio is around 3.5 to 3.8 seconds long, with some outliers around 4.5 seconds to a maximum of 5.27 seconds. We decided to listen to some of the audio with outlier durations but nothing out of the ordinary was found.

Distribution of Audio Length	**Mean and SD of Audio by Emotion**
	
Min Duration	**Max Duration**
2.94 seconds	5.27 seconds

Finally, looking into the Waveform, Log-Mel Spectrogram and MFCC of a sample audio clip from the dataset. This will be discussed further later during the data pre-processing section (2.3).



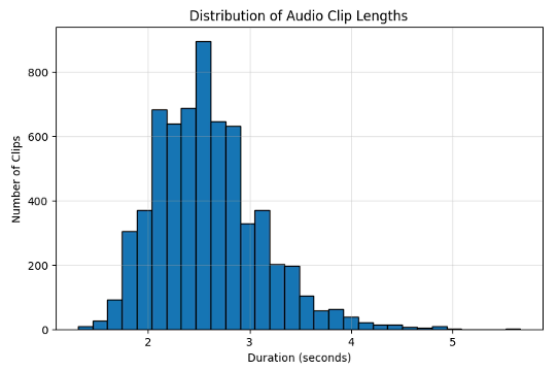
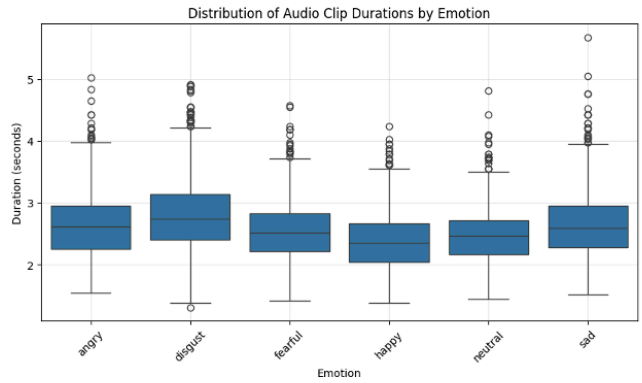
CREMA-D: A bigger dataset than RAVDESS, with a larger sample size. However, the audio recordings are not as structured and results in more data imbalances (e.g. gender imbalance). Furthermore, the crowd-sourcing emotion labelling approach taken may result in some accuracy issues.

The count of each label are:

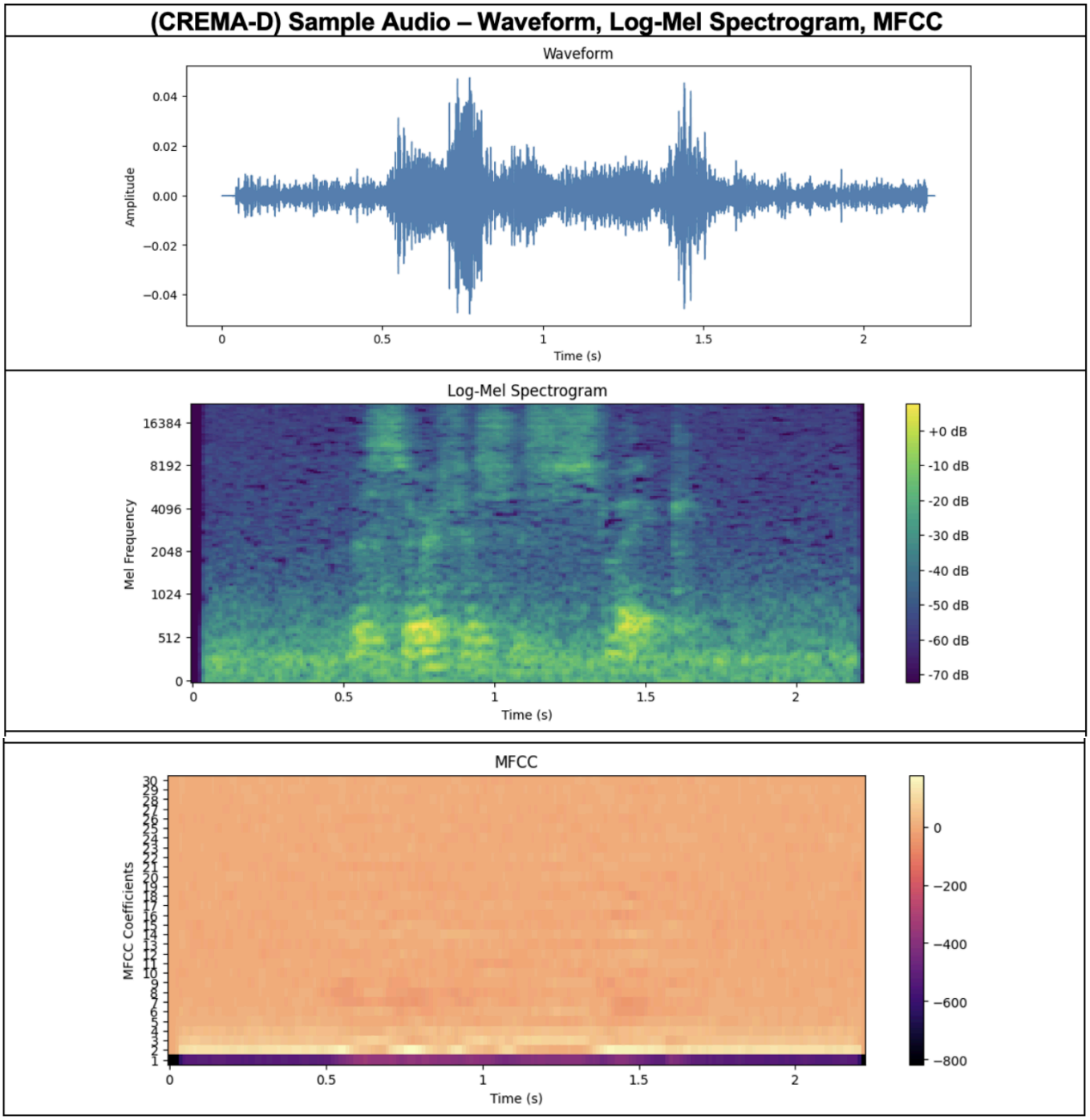
Emotions	Count of Each Emotion							
	Neutral	Calm	Happy	Sad	Angry	Fearful	Disgust	Surprised
Counts	934	1096	1093	1096	1096	1095	934	1096

Table 2.2.2 CREAMA-D Emotion Counts

Plotting the distribution of the durations show us that most of the audio is around 2 to 3 seconds long, with the median around 2.6 seconds, and some outliers above roughly 3.5 to 4 seconds. Compared to RAVDESS, the median duration is more uniformed across the different emotions, with a larger range of durations. We decided to listen to some of the audio with outlier durations but again, nothing out of the ordinary was found.

Distribution of Audio Length	**Mean and SD of Audio by Emotion**
	
Min Duration	**Max Duration**
1.31 seconds	5.67 seconds

CREMA-D exhibits consistently noisier waveforms than RAVDESS, a trend confirmed through manual inspection. The noise is also reflected though less prominently in the spectrogram and MFCC representations.

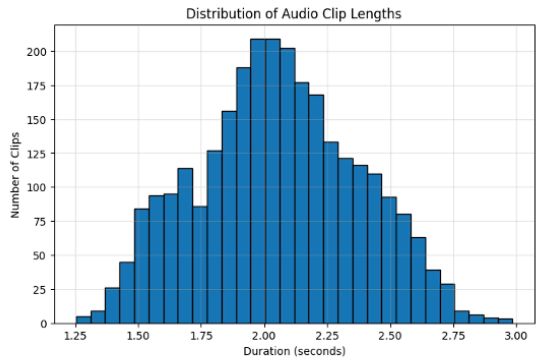
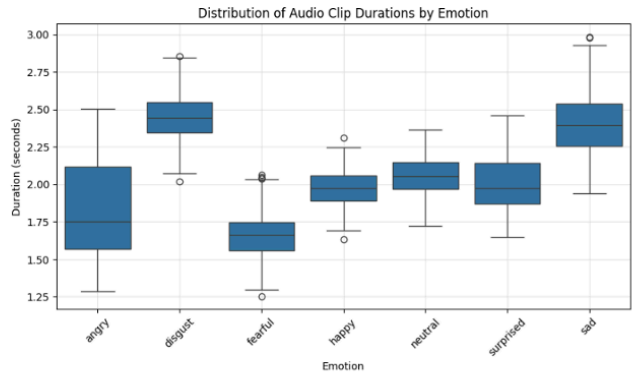


TESS: The data here is more abundant compared to RAVDESS but less than CREMA-D. However, the variation is limited in terms of different voice actors, intensity as well as by gender, since they were only recorded by 2 women. Looking into the data count we have:

	Count of Each Emotion							
Emotions	Neutral	Calm	Happy	Sad	Angry	Fearful	Disgust	Surprised
Counts	400	400	400	400	400	400	400	400

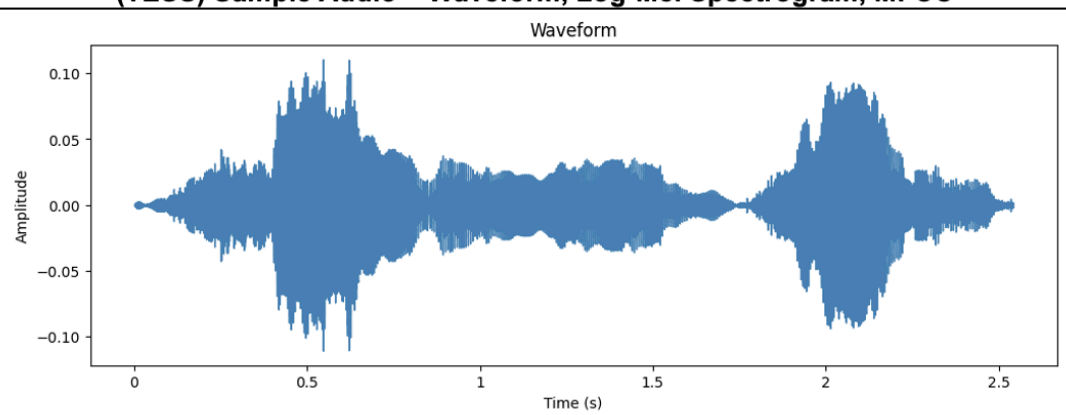
Table 2.2.3 TESS Emotion Counts

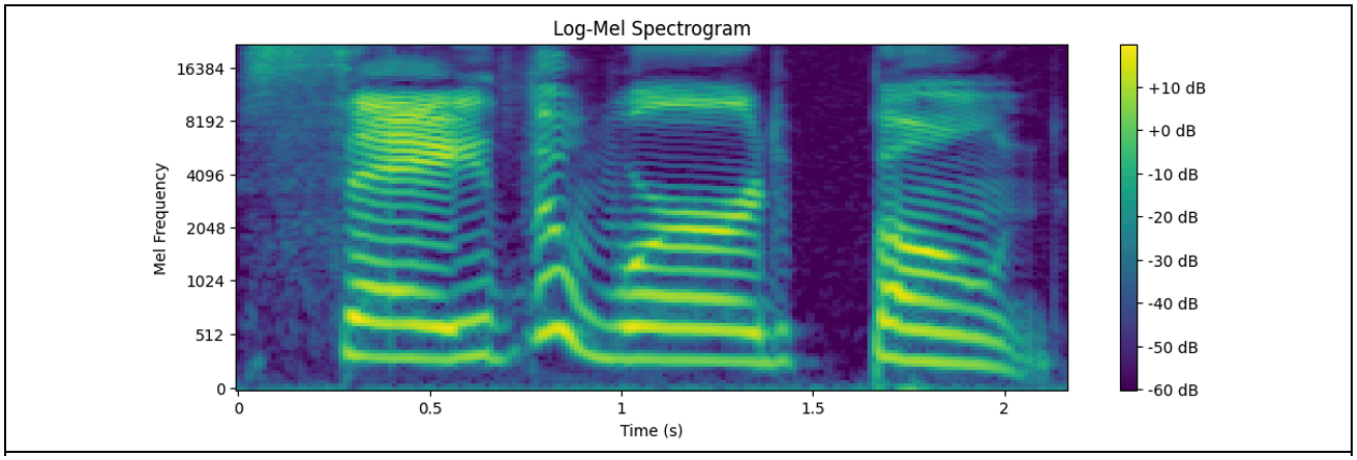
Plotting the duration distribution shows that TESS audio is far more uniform than both RAVDESS and CREMA-D. When examined by emotion class, the durations remain distinct with very few outliers, reflecting a controlled recording setup where speaking rate naturally differs across emotional states.

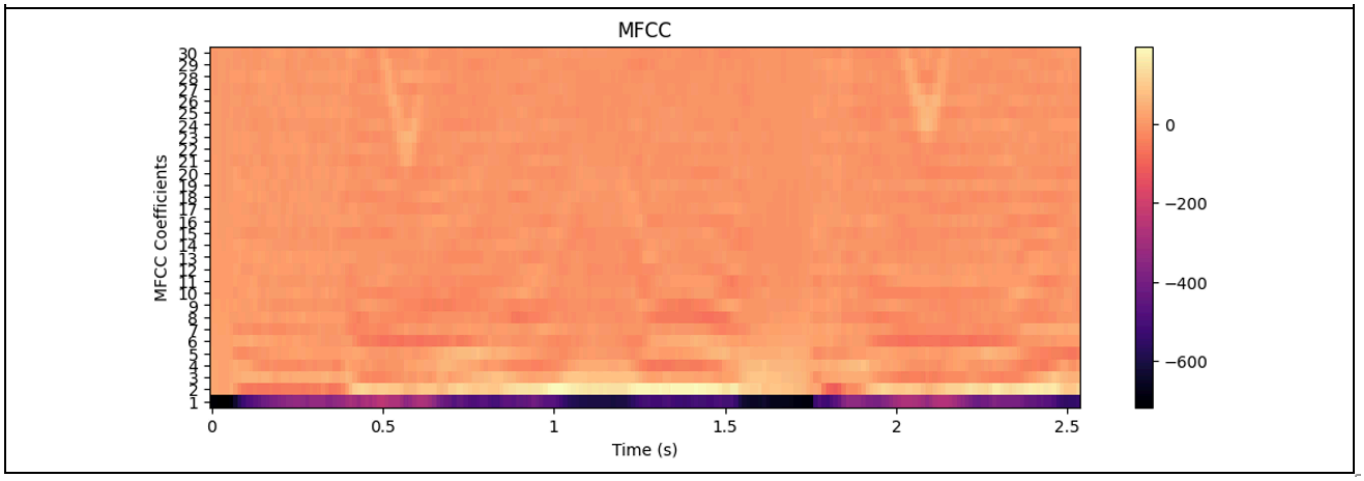
Distribution of Audio Length	**Mean and SD of Audio by Emotion**
	
Min Duration	**Max Duration**
1.25 seconds	2.98 seconds

Inspection of the waveforms shows consistent audio throughout each clip, with minimal silence at the start or end. The Log-Mel Spectrograms and MFCCs also reveal more varied tonal patterns than the previous datasets, suggesting more expressive and clearly articulated emotional recordings.

(TESS) Sample Audio – Waveform, Log-Mel Spectrogram, MFCC







2.3 Pre-processing

2.3.1 Data Cleaning

Each dataset was imported and reviewed separately before merging. Although RAVDESS, CREMA-D, and TESS share overlapping emotional categories, their label sets differ: RAVDESS contains 8 emotions, CREMA-D has 6, and TESS provides 7. To form a unified dataset, labels were standardised to seven final classes: neutral, happy, sad, angry, fearful, disgust, and surprised.

In RAVDESS, “neutral” and “calm” represent similar affective states; because CREMA-D and TESS do not include “calm,” it was merged into “neutral” to avoid class imbalance. CREMA-D does not contain “surprised,” but RAVDESS and TESS provide sufficient samples, so this class was retained to preserve emotional diversity. We also removed corrupted or zero-duration files identified in CREMA-D. After resolving inconsistencies, all remaining audio files were combined into a single dataset.

To address variable audio lengths, all clips were standardised using post-padding. Truncation was avoided as it risks discarding emotionally meaningful segments. Padding was applied at the end of each clip to preserve temporal ordering—critical for sequence models such as LSTMs—and to avoid injecting silence at the start, which could distort learned temporal patterns.

2.3.2 Feature Engineering, Feature Extraction, and Data Augmentation

Two commonly used audio features in speech analysis are the Log-Mel Spectrogram and MFCCs (Mel-Frequency Cepstral Coefficients). The Log-Mel Spectrogram is a 2D time–frequency representation that visually resembles an image, making it well-suited for convolutional neural networks (CNNs). MFCCs, on the other hand, produce a 1D vector of coefficients per frame, which can be stacked over time to form a compact 2D representation of the speech signal [3].

To determine which feature type to use, we trained a simple CNN–LSTM model and compared their validation accuracy. The table below summarises the results.

	Validation Accuracy Comparison
Feature	Accuracy
Log-Mel Spectrogram	57.3%
MFCC	68.5%

Table 2.3.2.1 Validation Accuracy Comparison

Given the higher baseline accuracy and the limited computational resources available, we proceeded with MFCCs as our primary feature representation. The overall MFCC extraction workflow (and the Log-Mel computation process) is illustrated in the figure below [4–7].

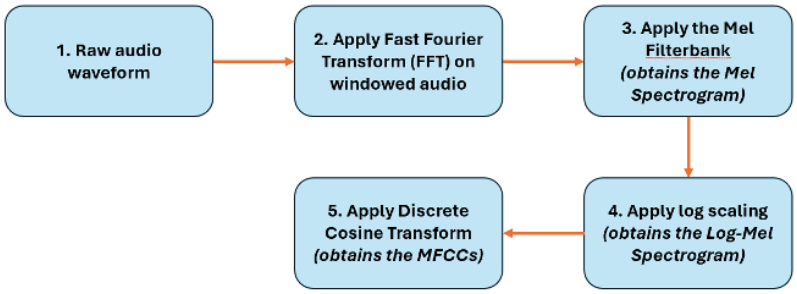


Figure 2.1. MFCC Extraction Workflow

1. The audio signal $x[n]$ is divided into short overlapping frames. Each frame is multiplied by a window function $w[n]$ to reduce spectral leakage. The general windowing operation is:

$$x_w[n] = x[n] \cdot w[n].$$

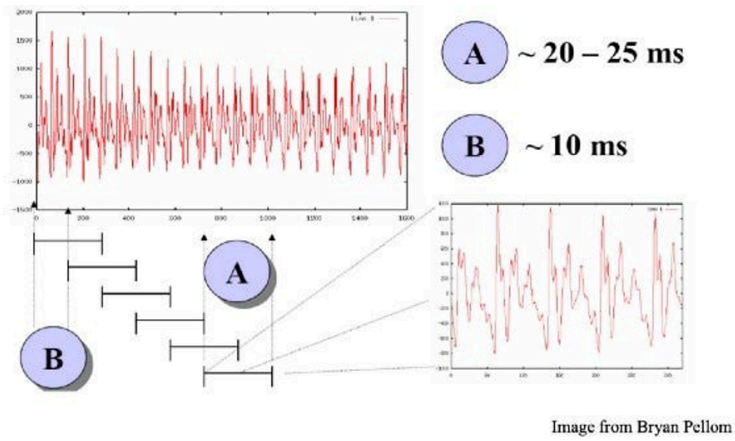


Figure 2.2. Dividing the audio into short frames

For this project, a Hann window was used:

$$w_H(n) = 0.5\left[1 - \cos\left(\frac{2\pi n}{N}\right)\right], \quad 0 \le n \le N-1,$$

where N is the frame length [8].

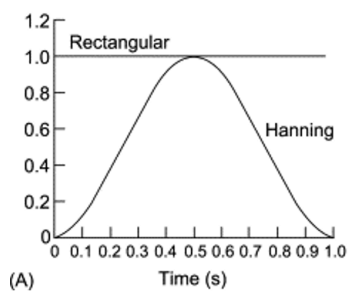


Figure 2.3. Applying a Hanning Window vs the default rectangular window

2. A Short-Time Fourier Transform (STFT) is performed, which applies an FFT to each windowed frame to obtain a time–frequency representation:

$$X(k,m) = \sum_{n=0}^{N-1} x_w[n + mH]\, e^{-j2\pi kn/N},$$

where

- k is the frequency bin index,
- m is the frame index,
- H is the hop length, and
- N is the FFT size.

The output is then squared to obtain the power spectrogram:

$$P(k, m) = |X(k, m)|^2.$$

3. A Mel filterbank is applied to map the linear frequency bins onto the Mel scale, yielding the Mel spectrogram:

$$S_{\text{mel}}(i, m) = \sum_{k=0}^{K-1} P(k, m)\, M_i(k),$$

where $M_i(k)$ is the i-th triangular Mel filter and K is the number of FFT frequency bins (see, e.g., https://apxml.com/courses/applied-speech-recognition/chapter-2-feature-extraction-for-speech/filter-banks-log-mel-spectrograms).

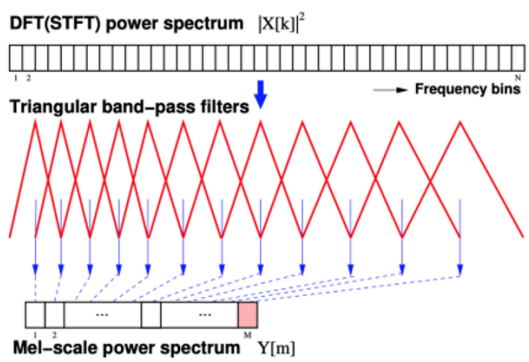


Figure 2.4. Mapping the power spectrum using the Mel Scale into the Mel Spectrum

4. A logarithm is applied to compress the dynamic range and better approximate human loudness perception, giving the Log-Mel spectrogram:

$$S_{\text{log-mel}}(i, m) = \log\left(S_{\text{mel}}(i, m) + \epsilon\right),$$

where ϵ is a small constant added for numerical stability.

5. To decorrelate the Mel energies and obtain a more compact representation, a Discrete Cosine Transform (DCT) is applied to each Log-Mel frame to compute the MFCCs:

$$\text{MFCC}_c(m) = \sum_{i=0}^{I-1} S_{\text{log-mel}}(i, m) \cos\left[\frac{\pi c}{I}\,(i + 0.5)\right],$$

where c is the MFCC coefficient index and I is the number of Mel bands. In addition to the MFCCs, we also extracted:

- **Delta (Δ) and Delta-Delta ($\Delta\Delta$) coefficients:** Capture the 1st and 2nd order temporal changes in MFCCs, which may be useful in emotion classification.
- **ZCR (Zero-Crossing Rate):** Measures how frequently the waveform crosses zero amplitude, which may be useful in determining articulation and manner of speech linked to emotion.
- **RMSE (Root Mean Square Energy):** Measures the energy of the speech signal, which may help distinguish between “strong” vs. “weak” emotions.

To allow the model to better generalize to unseen data, we also increased data variability and effective sample size by applying **three augmentation techniques** to the raw audio separately:

- **Additive Noise:** Low-level Gaussian noise was introduced to simulate background noise and recording imperfections without overpowering the speech, improving robustness.
- **Pitch Shifting:** Slightly alters the frequency of speech, simulating natural pitch variation across different speakers.
- **Time Stretching:** Modifies the speech rate to represent differences in speaking pace across emotions.

After several rounds of tuning hyperparameters, feature engineering, data extraction, data augmentation, and model evaluation—guided by recommendations from multiple works in the literature—we finalized the following feature extraction parameters:

Parameter	Purpose	Chosen Value
SR	Number of samples in 1 second of audio. For example, a sampling rate of 44.1kHz means 44,100 samples per second.	44100
N_MFCC	Number of coefficients from the Mel-Frequency Cepstral analysis.	30
N_FFT	Number of samples included in each FFT window.	2048
HOP_LENGTH	The distance (in samples) between successive windows. A window size of 2048 with hop length of 512 means 75% overlap.	512
WINDOW_TYPE	Type of taper applied to each frame before FFT to minimize spectral leakage.	hann

Table 2.3.2.2 Final Parameter Selection

2.3.3 Dataset Splitting

Finally, the dataset was split into train/validation/test sets (60/20/20) using stratified sampling to maintain proportional label distribution and reduce class imbalance during training.

Class	Train	Validation	Test	Total
Neutral	3892	1300	1296	6488
Happy	4052	1348	1352	6752
Sad	4044	1348	1348	6740
Angry	4052	1348	1352	6752
Fearful	4048	1352	1352	6752
Disgust	4052	1348	1348	6748
Surprised	1420	476	472	2368
Total	25560	8520	8520	42600

Table 2.3.3 Class Distribution in Datasets

3 Machine Learning Models

For computational efficiency, the models were first trained only on the non-augmented feature set (Stage 1), consisting of 30 MFCC coefficients and their first- and second-order temporal derivatives (Δ, $\Delta\Delta$) extracted from the combined RAVDESS, CREMA-D, and TESS datasets. The top 3 performing models from Stage 1 will be selected for Stage 2 where retraining with augmented features is performed.

3.1 Logistics Regression

Logistic Regression was implemented as a baseline classical model for speech emotion recognition (SER). It was selected due to its simplicity, interpretability, and ability to provide insights into which acoustic features best differentiate emotional expressions. As a linear classifier, it offers a foundational benchmark against which more complex models can be compared.

To support multi-class classification over the seven final emotion labels, we used a multinomial Logistic Regression formulation using the nnet::multinom() function in R. An L2 penalty (ridge regularization) was applied to reduce overfitting and prevent overly large weight coefficients dominating the decision boundary. The model was trained using maximum likelihood optimization with up to 1,000 iterations to ensure convergence.

During training, we utilised the previously defined 60/20/20 train-validation-test split, with stratified sampling for emotion balance. Class weights were also incorporated to reduce the influence of over-represented emotions and improve minority-class sensitivity.

As a purely linear model, Logistic Regression cannot fully capture nonlinear relationships and overlapping acoustic patterns between certain emotions. Therefore, while effective as an interpretable baseline, its performance establishes a point of comparison motivating the exploration of richer nonlinear models such as Support Vector Machines and ensemble learning methods in subsequent subsections.

3.2 Support Vector Machines (SVMs)

Support Vector Machines (SVMs) were adopted as a nonlinear classical model to improve classification performance beyond the linear decision boundaries learned by Logistic Regression. SVMs are margin-based classifiers that aim to determine the optimal separating hyperplane between classes by maximising the margin between support vectors, the most informative data points near the decision boundary. For binary classification, this is represented by the following convex optimisation problem:

$$\begin{aligned} \min_{w,\,b} \quad & \frac{1}{2}\|w\|^2 \\ \text{subject to:} \quad & y_i\left(w^\top x_i + b\right) \geq 1, \quad \forall i \end{aligned}$$

where w is the weight vector, b is the bias term, and y_i is the class label corresponding to the data point x_i.

Since this task involves seven emotional classes, a One-vs-One (OvO) extension was applied. To better separate overlapping emotional cues (e.g., between fear and surprise), multiple kernel functions were evaluated to allow flexible nonlinear mappings into higher-dimensional feature spaces:

- **Linear kernel:**

$$K(x, x') = x^\top x'$$

- **Polynomial kernel:**

$$K(x, x') = (x^\top x' + c)^d$$

where c is a constant and d is the degree of the polynomial.

- **Radial Basis Function (RBF) kernel:**

$$K(x, x') = \exp\left(-\frac{\|x - x'\|^2}{2\sigma^2}\right)$$

where σ is the width of the Gaussian.

Kernel	Parameter	Values Tested
Linear Kernel	Regularization	0.1, 1, 10
Polynomial Kernel	Degree	2, 3, 4
	Scale	0.001, 0.01, 0.1
	Regularization	0.1, 1, 10, 50
RBF Kernel	Sigma	0.001, 0.01, 0.1
	Regularization	0.1, 1, 10, 100

Table 3.2 Kernel Selection

All models were trained using the non-augmented MFCC feature set. Standardisation was applied to ensure equal weighting across coefficients, and class weighting helped address emotion imbalance. Hyperparameters were tuned via 3-fold cross-validation and grid search.

Across kernel comparisons, the RBF SVM consistently demonstrated superior ability to capture nonlinear structure in the acoustic feature space. Its Gaussian mapping enables emotion categories with subtle spectral temporal differences to become linearly separable in feature space making it well-suited for SER tasks.

Therefore, while multiple variants were initially explored, only the RBF SVM was retained as the final candidate SVM model for evaluation and comparison in Section 6. As mentioned, training was conducted only on non-augmented data first (Stage 1).

3.3 Random Forest and XGBoost

Ensemble tree-based models were incorporated into this study to provide stronger nonlinear modeling capacity than Logistic Regression and greater interpretability than deep learning approaches. Both Random Forest (RF) and eXtreme Gradient Boosting (XGBoost) operate by combining multiple decision trees; however, they differ in how trees are constructed and optimised. These models are widely used in SER tasks as they handle high-dimensional acoustic features effectively and are robust to noise and overfitting.

3.3.1 Random Forest (RF)

Random Forest builds a collection of decision trees and aggregates their outputs through majority voting. It introduces randomness in two ways:

1) Bootstrap Aggregating (Bagging): Each tree is trained on a random subset of the data $\rightarrow$ lowers variance.

2) Random Feature Selection: At each split, only a random subset of features is considered $\rightarrow$ increases model diversity.

The Gini impurity is used as the split criterion:

$$\text{Gini}(t) = 1 - \sum_{i=1}^{C} p_i^2$$

where p_i is the proportion of samples belonging to class i at node t, and C is the number of emotion classes.

The final parameters chosen for the Random Forest (RF) classifier are shown here.

Hyperparameters	**Values**
mtry	8
min.node.size	1
splitrule	Gini

Table 3.3.1 RF Hyperparameter Selection

As per the study's two-stage evaluation pipeline, the RF model was trained only in Stage 1.

3.3.2 eXtreme Gradient Boosting (XGBoost)

XGBoost improves upon standard boosting by building trees sequentially, where each new tree corrects the residual errors of previous trees. The model optimises an objective function consisting of a loss term and a regularisation term:

$$\text{Obj} = \sum_{i=1}^{n} l(y_i, \hat{y}_i^{(t)}) \; + \; \sum_{k=1}^{t} \Omega(f_k)$$

The regularisation component penalises overly complex trees:

$$\Omega(f_k) = \gamma T \; + \; \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2$$

XGBoost also incorporates **second-order optimisation** using gradients g_i and Hessians h_i:

$$L(f) = \sum_{i=1}^{n} \left[g_i f(x_i) \; + \; \frac{1}{2} h_i f^2(x_i) \right] + \Omega(f)$$

This improves convergence efficiency and yields strong performance, especially on imbalanced SER data.

Hyperparameters	Values
colsample_bytree	0.8
eta (learning rate)	0.1
gamma	0
min_child_weight	1
max_depth	8
nrounds	200
subsample	0.8

Table 3.3.2 XGBoost Hyperparameter Selection

As per the study's two-stage evaluation pipeline, the XGBoost model was trained first in Stage 1, and moved on to Stage 2 with its promising results.

3.4 Convolutional Neural Networks (CNN)

CNNs were implemented to automatically learn acoustic representations directly from MFCC feature maps, allowing emotional cues to be captured from localised time–frequency patterns. Unlike classical models such as SVMs or Random Forests, which rely on manually engineered features, CNNs can extract hierarchical patterns related to vocal pitch variation, spectral dynamics, and energy contours.

Several input channel variants were explored:

- **MFCC only**
- **MFCC +** Δ
- **MFCC +** $\Delta + \Delta\Delta$

Based on validation results, the baseline CNN adopted **MFCC +** Δ, introducing first-order temporal variation while avoiding excessive parameter growth.

The CNN architecture consisted of three **convolution** → **batch normalisation** → **ReLU** → **max-pooling** blocks, with neurons increasing as:

$$32 \rightarrow 64 \rightarrow 128$$

Convolutional layers learn short-range spectral correlations, while max-pooling along the temporal dimension reduces redundant frames and introduces invariance to speaking-rate differences. A fully connected layer followed by a SoftMax activation was used for seven-class emotion classification.

To mitigate overfitting due to limited speaker diversity in RAVDESS and TESS, several regularisation strategies were applied:

- **Dropout:** rate = 0.5
- **L2 weight decay:** 1×10^{-4}
- **Label smoothing:** $\varepsilon = 0.05$
- **Early stopping** based on validation loss

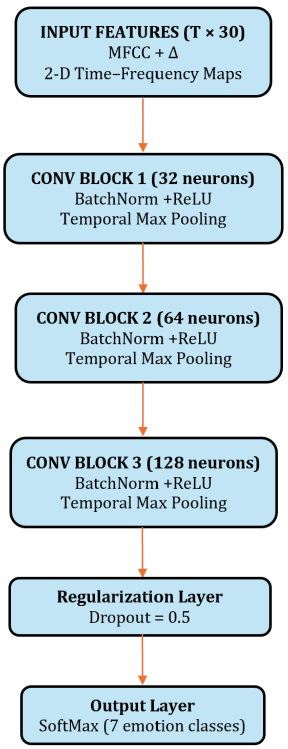


Figure 3.1. CNN Workflow

Training used the Adam optimiser (learning rate = 0.001), with a batch size of 32 and a maximum of 100 epochs. Inputs were standardised using z-score normalisation from the training distribution.

As per the study's two-stage evaluation pipeline, the CNN model was trained first in Stage 1. Whether the model progresses into Stage 2 augmented retraining will be determined based on its performance relative to other models.

3.5 Bidirectional LSTMs (BiLSTM)

Recurrent neural networks are well-suited to speech emotion recognition (SER) as emotion is expressed dynamically over time rather than within isolated frames [9]. Long Short-Term Memory (LSTM) networks improve upon simple RNNs through gating mechanisms that retain or forget information, enabling robust modelling of long-range vocal patterns such as stress shifts and tonal changes. A Bidirectional LSTM (BiLSTM) extends this architecture by processing MFCC sequences both forward and backward, thereby learning contextual acoustic dependencies that appear before and after each speech frame, a valuable property in detecting subtle emotional contours.

The model receives non-augmented MFCC-based features shaped as (samples × timesteps × feature channels). Input normalisation was also applied to ensure comparable feature scaling.

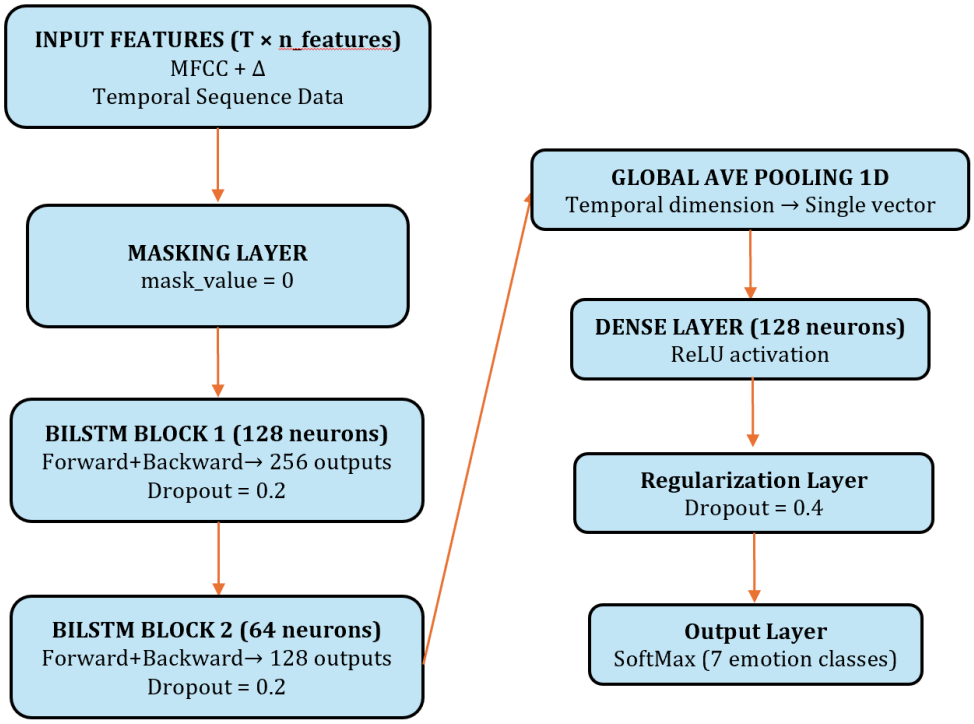


Figure 3.2. BiLSTM Workflow

This architecture allows:

- Extraction of **bidirectional temporal cues** relevant to affect
- Improved **invariance to speaking-rate variation**
- Learning of **global emotional prosody** beyond local spectral cues

To enhance generalisation, regularisation techniques included **dropout**, **recurrent dropout**, and **early stopping**. A **class-weighted loss** was applied to compensate for lower representation of minority emotions (e.g., surprise), ensuring proportional contribution during optimisation.

Training used the **Adam optimiser** with an initial learning rate of 3×10^{-4} and **ReduceLROnPlateau**, which adaptively decreases the learning rate when validation performance stagnates.

In Stage 1, the BiLSTM was trained using only non-augmented MFCC sequences to benchmark performance against classical and CNN models. Owing to its superior ability to capture temporal emotion dynamics, BiLSTM was selected as one of the top 3 models to continue into Stage 2 with augmented-data.

3.6 CNN–Long Short-Term Memory (CNN-LSTM)

To simultaneously capture local spectral changes and evolving emotional prosody over time, a hybrid CNN-LSTM architecture was implemented. This model combines the strengths of CNNs, which learn spatially structured features from MFCC representations, with BiLSTMs that capture long-range temporal dependencies. This aligns closely with how emotions manifest in speech, through both immediate spectral fluctuations and gradual shifts in tone, intensity, and rhythm.

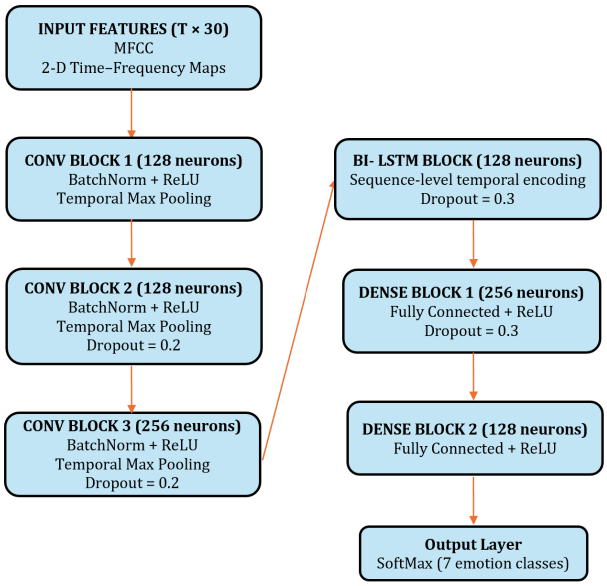


Figure 3.3. CNN-LSTM Workflow

The input consisted of non-augmented MFCC features arranged into 2-D time-frequency maps. Local patterns such as harmonics, formant trajectories, and sharp energy transitions were extracted through three sequential convolutional blocks.

Each block applied batch normalisation, ReLU activation, and max-pooling along the temporal axis, progressively compressing redundant frames while preserving frequency-selective detail. Dropout with rate 0.2 was applied after the second and third convolutional blocks to reduce overfitting.

The resulting feature sequence was fed directly into a 128-neuron Bidirectional LSTM layer, which produced a fixed-length temporal embedding representing the entire utterance; an additional dropout layer (rate = 0.3) regularised this representation. This embedding was then passed through two fully connected layers with 256 and 128 neurons (ReLU activation), separated by dropout (rate = 0.3), before a final SoftMax layer produced class probabilities over the seven emotion categories.

The network was trained using categorical cross-entropy and the Adam optimiser, with performance monitored on a held-out validation set. Within the two-stage experimental design, the CNN–LSTM was first evaluated on non-augmented features to benchmark its temporal–spectral modelling effectiveness. With strong performance, it was selected as one of the top three models to proceed to Stage-2 retraining with augmented audio and K-fold cross-validation.

4 Evaluation Metrics

Multiple evaluation metrics were used to accurately assess emotion classification performance under imbalanced class distributions. The following metrics were computed on the held-out test set for all models.

- **Accuracy**
  Proportion of correctly predicted samples across all emotion categories.
- **Precision**
  Fraction of predicted instances of a given emotion that are correct. This is particularly important in healthcare settings to avoid false alarms and unnecessary escalation.

- **Recall**
  Fraction of true emotional expressions correctly identified. High recall is critical for SSD detection, where missing subtle signs of distress may delay intervention.
- **Macro F1-Score**
  Harmonic mean of precision and recall averaged equally across classes. This prevents majority classes from dominating model evaluation and provides a more balanced view of performance.
- **Confusion Matrix**
  Visualises misclassification patterns between emotions (e.g., fear vs surprise), helping to interpret where acoustic overlaps occur and which emotions are systematically confused.

These metrics collectively provide not only an overall measure of predictive accuracy but also detailed insight into which emotions are harder to detect, a key requirement for deployment in real-world elderly monitoring environments.

5 Results and Analysis

This section presents the experimental results from the two-stage benchmarking pipeline: (i) baseline model evaluation using non-augmented speech data, followed by (ii) retraining of top-performing models using an augmented dataset to enhance generalisability. Performance is evaluated using accuracy, precision, recall, and macro-averaged F1-score to account for class imbalance.

5.1 Stage 1: Baseline Evaluation (Non-Augmented Data)

All six candidate models were first trained on the original dataset to establish baseline performance while minimising computational cost. Table 5.1.1 summarises accuracy and macro-F1 scores on the test set.

5.1.1 Overall Model Comparison

Model	Accuracy	Macro-F1 Score
Logistic Regression	0.55	0.56
RBF SVM	0.44	0.42
XGBoost	0.61	0.60
CNN	0.54	0.70
BiLSTM	0.70	0.72
CNN-LSTM	0.67	0.64

Table 5.1.1: Accuracy and Macro-F1 Score

BiLSTM achieved the highest performance across both metrics, demonstrating the benefit of bidirectional temporal modelling over purely spectral architectures.

5.1.2 Recall by Emotion

Emotion	LR	XGB	SVM	CNN	BiLSTM	CNN-LSTM
Neutral	0.60	0.68	0.76	0.80	0.85	0.82
Happy	0.46	0.51	0.61	0.31	0.66	0.70
Sad	0.65	0.69	0.27	0.60	0.69	0.61
Angry	0.73	0.76	0.65	0.81	0.81	0.68
Fear	0.42	0.44	0.19	0.59	0.66	0.64
Disgust	0.41	0.49	0.11	0.26	0.57	0.53
Surprise	0.70	0.84	0.45	0.19	0.95	0.95

Table 5.1.2: Recall by Emotion

Consistently strong recall across negative emotions is crucial for early SSD detection.

5.1.3 Precision by Emotion

Emotion	LR	XGB	SVM	CNN	BiLSTM	CNN-LSTM
Neutral	0.54	0.56	0.39	0.44	0.70	0.68
Happy	0.51	0.55	0.38	0.43	0.51	0.67
Sad	0.53	0.59	0.58	0.56	0.72	0.65
Angry	0.63	0.69	0.61	0.64	0.88	0.79
Fear	0.54	0.60	0.25	0.54	0.63	0.69
Disgust	0.50	0.58	0.37	0.77	0.71	0.76
Surprise	0.63	0.80	0.65	1.00	0.77	0.94

Table 5.1.3: Precision by Emotion

Both Bi-LSTM and CNN-LSTM improves precision dramatically for fear, disgust, surprise which are emotions associated with distress.

5.1.4 Model Selection for Stage 2 Retraining

The following criteria were used to determine which models should be advanced:

- Macro-level performance (not only accuracy)
- Recall on distress-related emotions
- Generalisation potential for real-world deployment

Based on these criteria, CNN-LSTM, BiLSTM, and XGBoost were selected for Stage 2 training.

5.2 Stage 2: Performance with Augmented Data

After initial screening, the top three models were retrained using the augmented dataset to improve robustness against noise and speaker variability. Their final evaluation on the held-out test set is presented below.

5.2.1 Overall Model Comparison (Augmented Data)

Model	Accuracy	Macro-F1 Score
XGBoost	0.61	0.59
BiLSTM	0.72	0.74
CNN-LSTM	**0.73**	**0.75**

Table 5.2.1: Accuracy and Macro-F1 Score (Augmented Data)

CNN-LSTM demonstrates the strongest overall performance, improving both accuracy and macro-F1 following augmentation.

5.2.2 Recall by Emotion (Augmented Data)

Emotion	XGBoost	BiLSTM	CNN-LSTM
Neutral	0.68	0.86	**0.82**
Happy	0.53	0.66	**0.65**
Sad	0.68	0.64	**0.73**
Angry	0.76	0.82	**0.87**
Fear	0.47	0.60	**0.67**
Disgust	0.49	0.55	**0.67**
Surprise	0.84	0.93	**0.96**

Table 5.2.2: Recall by Emotion (Augmented Data)

5.2.3 Precision by Emotion (Augmented Data)

Emotion	XGBoost	BiLSTM	CNN-LSTM
Neutral	0.56	0.66	**0.75**
Happy	0.56	0.70	**0.77**
Sad	0.60	0.68	**0.59**
Angry	0.71	0.78	**0.8**
Fear	0.62	0.67	**0.75**
Disgust	0.54	0.68	**0.76**
Surprise	0.83	0.91	**0.94**

Table 5.2.3: Precision by Emotion (Augmented Data)

CNN-LSTM is consistently more accurate in identifying clinically relevant negative emotional states.

5.3 Best Model – CNN-LSTM

Across both training stages, the CNN-LSTM demonstrated the strongest overall performance, achieving the highest accuracy and macro-F1 score while maintaining balanced recall across distress-related emotions. To better understand remaining weaknesses, a confusion matrix for the augmented-data test result is shown in Figure 5.1.

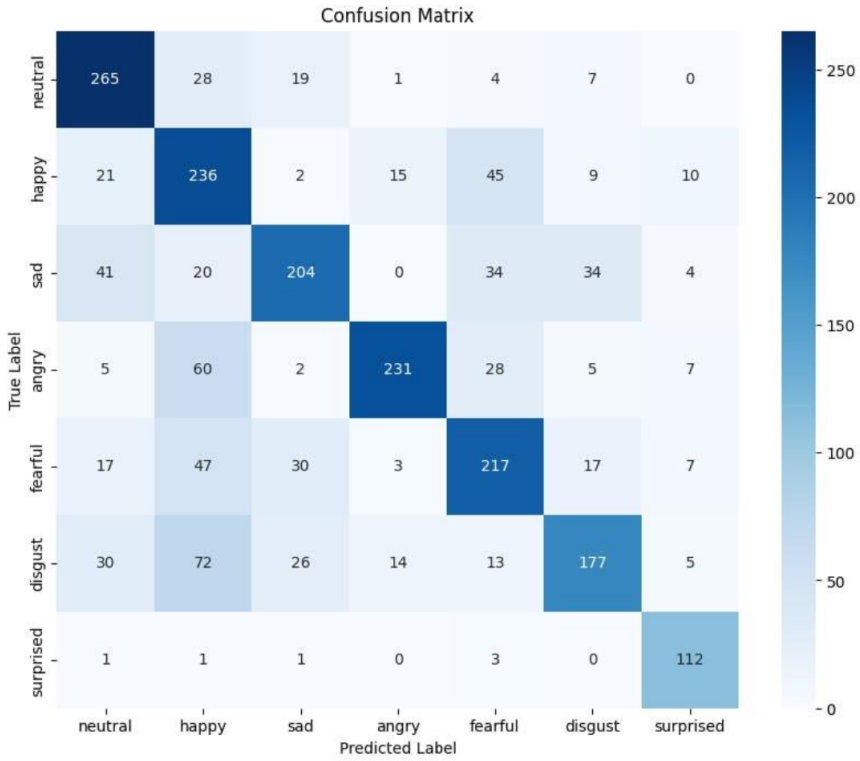


Figure 5.1: Confusion Matrix for CNN-LSTM (Augmented Test Set)

The model achieves strong diagonal dominance, particularly for angry and surprised, indicating reliable separation of high-arousal emotions. Most misclassifications occur between sad, fearful, and disgust, which share low-energy negative affect and overlapping acoustic patterns. Occasional confusion between happy and neutral is expected, as more subtle positive affect can be acoustically like flat or mildly expressive speech.

With continuous conversational input and noise-robust training, the CNN-LSTM is well-positioned for deployment in passive emotional-health monitoring systems.

6 Discussion

This project benchmarked a range of classical and deep-learning models for speech-emotion recognition (SER) using publicly available acted-speech datasets. Rather than targeting a specific clinical population, the study aimed to evaluate how well different architectures can model emotional cues in speech under controlled conditions, and to identify which approaches show the strongest potential for future real-world applications. Across both experimental stages, the CNN–LSTM consistently achieved the best overall performance, benefiting from its ability to learn both local spectral patterns and long-range temporal dependencies. These results align with broader SER research showing that hybrid temporal–spectral models outperform purely convolutional or recurrent networks, particularly for subtle or low-arousal emotions [10, 11].

Beyond accuracy, SER systems offer several practical advantages that motivate their use in downstream applications such as conversational agents, wellbeing-monitoring interfaces, and human–computer interaction:

- **Passive and continuous monitoring** voice-based cues can be captured during natural interaction without requiring additional user effort.
- **Scalability and accessibility** SER can be deployed on a wide range of devices, from mobile apps to embedded systems.
- **Multi-domain applicability** potential use cases include customer-service analytics, wellbeing tracking, social robotics, and adaptive learning systems.

However, significant challenges must be addressed before SER models can be deployed in real environments:

- **Data mismatch between acted and natural speech**

  The datasets used in this study contain clearly articulated, exaggerated emotional expressions, which differ substantially from spontaneous speech. This gap is widely documented in SER literature and remains a major barrier to generalisation [12].
- **Variability across speakers, accents, and environments**

  Acoustic differences caused by recording conditions, microphones, background noise, and speaker identity can strongly affect model robustness [13].
- **Interpretability and trust**

  Deep models such as CNN-LSTMs provide high accuracy but limited transparency. Explainability tools (e.g., saliency over MFCCs, attention heatmaps) will be essential for domains where decision accountability matters.
- **Ethical and privacy considerations**

  Voice data is biometric information and must be collected, stored, and processed under strict consent and governance frameworks [14].

7 Conclusion

This project benchmarked a range of classical and deep-learning approaches for speech-emotion recognition (SER) using three publicly available acted-speech datasets. By evaluating models across a two-stage pipeline—including baseline training and augmented-data retraining—the study provides a clearer understanding of how different architectures handle spectral and temporal cues in emotional audio.

Across all experiments, the CNN–LSTM model demonstrated the strongest and most consistent performance, reflecting the advantages of combining convolutional feature extraction with recurrent temporal modeling. These findings reinforce trends reported in contemporary SER literature and highlight the effectiveness of hybrid architectures for tasks involving both short-term and long-range acoustic dependencies.

While the benchmarking results are promising, they also emphasise key limitations inherent in current SER research: acted datasets differ substantially from natural conversational speech, speaker variability remains a challenge, and deep models require greater interpretability for responsible real-world use. Addressing these issues will be essential for future work.

Potential next steps include expanding evaluation to spontaneous or in-the-wild emotional speech, incorporating pretrained or self-supervised audio representations, and developing explainability tools to make model behaviour more transparent. Additional robustness testing across accents, noise conditions, and demographic groups would further strengthen external validity.

Overall, this project contributes a structured and reproducible SER benchmark that can support future research on emotion-aware conversational systems, wellbeing analytics, and human–computer interaction. While not ready for deployment, the results offer a strong technical foundation for continued exploration and refinement of speech-based emotion modelling.

You may access all source code and data files here:
MH4510 Winners Report – Code & Data Repository

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